

求级数

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} \frac{(-1)^{i+j}}{i+j}$$

考虑到

$$\frac{(-1)^{i+j}}{i+j} = \int_0^{-1} x^{i+j} \mathrm{d}x$$

则有

$$\begin{aligned} &\int_0^{-1} \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} x^{i+j} \mathrm{d}x \\ &= \int_0^{-1} \frac{x}{(1-x)^2} \mathrm{d}x \\ &= \ln 2 - \frac{1}{2} \end{aligned}$$